

Context-Free Grammars In Class Examples and Extra Problems

1. Write a context-free grammar for the following languages.

(a) $L = \{a^n w w^R b^n : w \in \{a, b\}^*, n > 0\}$

$$\begin{aligned} S &\rightarrow aSb \mid aAb \\ A &\rightarrow aAa \mid bAb \mid \lambda \end{aligned}$$

(b) $L = \{a^n b^n : n \text{ is even}\}$

$$S \rightarrow aaSbb \mid \lambda$$

(c) $L = \{a^n b^n : n \text{ is odd}\}$

$$\begin{aligned} S &\rightarrow aAb \\ A &\rightarrow aaAbb \mid \lambda \end{aligned}$$

(d) $L = \{a^n b^m : n \neq m\}$

$$\begin{aligned} S &\rightarrow S_1 \mid S_2 \\ S_1 &\rightarrow aS_1b \mid S_1b \mid b \\ S_2 &\rightarrow aS_2b \mid aS_2 \mid a \end{aligned}$$

(e) $L = \{a^n b^m : 2n \leq m \leq 3n \text{ and } n \geq 0, m \geq 0\}$

$$S \rightarrow aSbb \mid aSbbb \mid \lambda$$

(f) $L = \{a^n b^m : n \leq m + 3\}$

$$\begin{aligned} S &\rightarrow aSb \mid A \\ A &\rightarrow Ab \mid a \mid aa \mid aaa \mid \lambda \end{aligned}$$

(g) $L = \{a^n b^m : n = m - 1\}$

$$S \rightarrow aSb \mid b$$

(h) $L = \{a^n b^m c^l : n \geq 0, m \geq 0, l \geq 0, l = m + n\}$

$$\begin{aligned} S &\rightarrow aSc \mid A \\ A &\rightarrow bAc \mid \lambda \end{aligned}$$

(i) $L = \{a^n b^m c^l : n \geq 0, m \geq 0, l \geq 0, n = m \text{ or } m \leq l\}$

$$S \rightarrow S_1 C \mid A S_2$$

$$S_1 \rightarrow a S_1 b \mid \lambda$$

$$S_2 \rightarrow b S_2 c \mid C$$

$$A \rightarrow a A \mid \lambda$$

$$C \rightarrow c C \mid \lambda$$

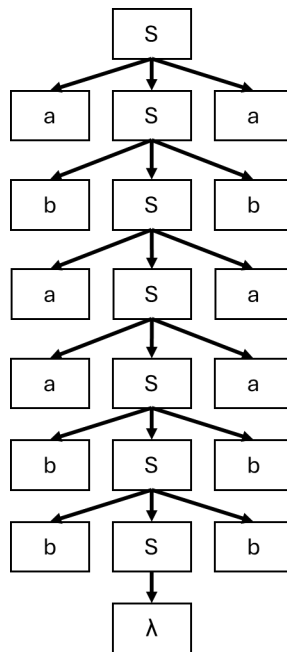
(j) $L = \{a^n b^m c^l : n \geq 0, m \geq 0, l \geq 0, l = 2m + n\}$

$$S \rightarrow a S c \mid A$$

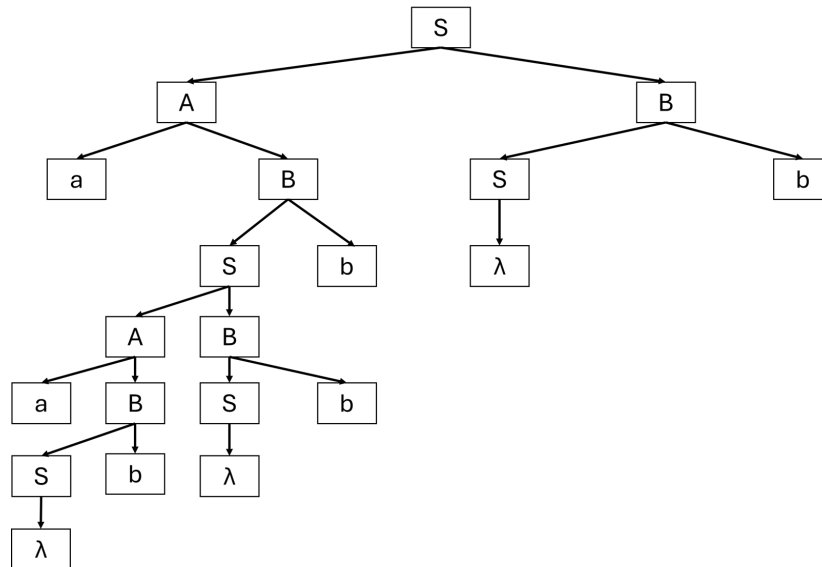
$$A \rightarrow b A c c \mid \lambda$$

2. Draw a derivation tree that creates the following sentence *abaabbaaba* from the below context-free grammar.

$$S \rightarrow a S a \mid b S b \mid \lambda$$



- $$B \rightarrow Sb$$



- $$B \rightarrow b \mid \lambda$$

$$B \rightarrow b \mid \lambda$$
$$B \rightarrow b$$

Step 3: Remove unit rules. $S_0 \rightarrow S$ and $S \rightarrow S$

$$\begin{aligned} S_0 &\rightarrow ASA \mid aB \mid a \mid SA \mid AS \\ S &\rightarrow ASA \mid aB \mid a \mid SA \mid AS \\ A &\rightarrow B \mid S \\ B &\rightarrow b \end{aligned}$$

Step 4: Remove unit rules. $A \rightarrow B$ and $A \rightarrow S$

$$\begin{aligned} S_0 &\rightarrow ASA \mid aB \mid a \mid SA \mid AS \\ S &\rightarrow ASA \mid aB \mid a \mid SA \mid AS \\ A &\rightarrow b \mid ASA \mid aB \mid a \mid SA \mid AS \\ B &\rightarrow b \end{aligned}$$

Step 5: Convert remaining rules into the proper form by adding additional variables.

$$\begin{aligned} S_0 &\rightarrow AA_1 \mid UB \mid a \mid SA \mid AS \\ S &\rightarrow AA_1 \mid UB \mid a \mid SA \mid AS \\ A &\rightarrow b \mid AA_1 \mid UB \mid a \mid SA \mid AS \\ A_1 &\rightarrow SA \\ U &\rightarrow a \\ B &\rightarrow b \end{aligned}$$

5. Convert the following context-free grammar into Chomsky Normal Form (CNF).

$$\begin{aligned} S &\rightarrow ABa \mid aB \\ A &\rightarrow aaB \\ B &\rightarrow Ac \end{aligned}$$

There are no λ rules or unit rules. All we have to do is put it in proper form by adding additional variables.

$$\begin{aligned} S &\rightarrow ABB_a \mid B_aB \\ A &\rightarrow B_aB_aB \\ B &\rightarrow AB_c \\ B_a &\rightarrow a \\ B_b &\rightarrow b \\ B_c &\rightarrow c \end{aligned}$$

Add more variables...

$$\begin{aligned}
 S &\rightarrow AC \mid B_a B \\
 A &\rightarrow DB \\
 B &\rightarrow AB_c \\
 B_a &\rightarrow a \\
 B_b &\rightarrow b \\
 B_c &\rightarrow c \\
 C &\rightarrow BB_a \\
 D &\rightarrow B_a B_a
 \end{aligned}$$

6. Construct a right-linear grammar for the strings that are generated by the regular expression $r = (aab^*ab)^*$

$$\begin{aligned}
 S &\rightarrow aaA \mid \lambda \\
 A &\rightarrow bA \mid abS
 \end{aligned}$$

7. Construct a regular grammar that generates the language $L = \{w \in \{a, b\}^* : n_a(w) + n_b(w) \text{ is even}\}$

In order to solve this, we need to keep track of the number of a's and b's we have. We can make special variables that represents if the number of a's and b's being even/odd.

A will represent when the number of a's is odd and number of b's is even.
 B will represent when the number of a's is even and number of b's is odd.
 C will represent when the number of a's is even and number of b's is even.
 D will represent when the number of a's is odd and number of b's is odd.

$$\begin{aligned}
 S &\rightarrow aA \mid bB \mid \lambda \\
 A &\rightarrow aC \mid bD \\
 B &\rightarrow aD \mid bC \\
 C &\rightarrow aA \mid bB \mid \lambda \\
 D &\rightarrow aB \mid bA \mid \lambda
 \end{aligned}$$

8. Convert the following context-free grammar into Chomsky Normal Form (CNF).

$$\begin{aligned} S &\rightarrow ab \mid Ab \mid B \\ A &\rightarrow C \mid \lambda \mid baB \mid B \\ B &\rightarrow acBc \mid bbCBa \mid aACB \mid ABA \mid bc \\ C &\rightarrow a \mid bac \mid CA \mid \lambda \end{aligned}$$

Step 1: Create new Start Variable S_0 .

$$\begin{aligned} S_0 &\rightarrow S \\ S &\rightarrow ab \mid Ab \mid B \\ A &\rightarrow C \mid \lambda \mid baB \mid B \\ B &\rightarrow acBc \mid bbCBa \mid aACB \mid ABA \mid bc \\ C &\rightarrow a \mid bac \mid CA \mid \lambda \end{aligned}$$

Step 2: Remove λ -rules. $A \rightarrow \lambda$

$$\begin{aligned} S_0 &\rightarrow S \\ S &\rightarrow ab \mid Ab \mid B \mid b \\ A &\rightarrow C \mid baB \mid B \\ B &\rightarrow acBc \mid bbCBa \mid aACB \mid ABA \mid bc \mid aCB \mid AB \mid BA \mid B \\ C &\rightarrow a \mid bac \mid CA \mid \lambda \mid C \end{aligned}$$

Step 3: Remove λ -rules. $C \rightarrow \lambda$

$$\begin{aligned} S_0 &\rightarrow S \\ S &\rightarrow ab \mid Ab \mid B \mid b \\ A &\rightarrow C \mid baB \mid B \\ B &\rightarrow acBc \mid bbCBa \mid aACB \mid ABA \mid bc \mid aCB \mid AB \mid BA \mid B \mid bbBa \mid aAB \mid aB \\ C &\rightarrow a \mid bac \mid CA \mid C \mid A \end{aligned}$$

Step 4: Remove Unit rules. $S_0 \rightarrow S$

$$\begin{aligned} S_0 &\rightarrow ab \mid Ab \mid B \mid b \\ S &\rightarrow ab \mid Ab \mid B \mid b \\ A &\rightarrow C \mid baB \mid B \\ B &\rightarrow acBc \mid bbCBa \mid aACB \mid ABA \mid bc \mid aCB \mid AB \mid BA \mid B \mid bbBa \mid aAB \mid aB \\ C &\rightarrow a \mid bac \mid CA \mid C \mid A \end{aligned}$$

Step 5: Remove Unit rules. $S_0 \rightarrow B$ and $S \rightarrow B$.

$$\begin{aligned} S_0 &\rightarrow ab \mid Ab \mid b \mid acBc \mid bbCBa \mid aACB \mid ABA \mid bc \mid aCB \mid AB \mid BA \mid bbBa \mid aAB \mid aB \\ S &\rightarrow ab \mid Ab \mid b \mid acBc \mid bbCBa \mid aACB \mid ABA \mid bc \mid aCB \mid AB \mid BA \mid bbBa \mid aAB \mid aB \\ A &\rightarrow C \mid baB \mid B \\ B &\rightarrow acBc \mid bbCBa \mid aACB \mid ABA \mid bc \mid aCB \mid AB \mid BA \mid B \mid bbBa \mid aAB \mid aB \\ C &\rightarrow a \mid bac \mid CA \mid C \mid A \end{aligned}$$

Step 6: Remove Unit rules. $A \rightarrow C, A \rightarrow B$, and $B \rightarrow B$.

$$\begin{aligned} S_0 &\rightarrow ab \mid Ab \mid b \mid acBc \mid bbCBa \mid aACB \mid ABA \mid bc \mid aCB \mid AB \mid BA \mid bbBa \mid aAB \mid aB \\ S &\rightarrow ab \mid Ab \mid b \mid acBc \mid bbCBa \mid aACB \mid ABA \mid bc \mid aCB \mid AB \mid BA \mid bbBa \mid aAB \mid aB \\ A &\rightarrow baB \mid a \mid bac \mid CA \mid acBc \mid bbCBa \mid aACB \mid ABA \mid bc \mid aCB \mid AB \mid BA \mid bbBa \mid aAB \mid aB \\ B &\rightarrow acBc \mid bbCBa \mid aACB \mid ABA \mid bc \mid aCB \mid AB \mid BA \mid bbBa \mid aAB \mid aB \\ C &\rightarrow a \mid bac \mid CA \mid C \mid A \end{aligned}$$

Step 7: Remove Unit rules. $C \rightarrow A$ and $C \rightarrow C$.

$$\begin{aligned} S_0 &\rightarrow ab \mid Ab \mid b \mid acBc \mid bbCBa \mid aACB \mid ABA \mid bc \mid aCB \mid AB \mid BA \mid bbBa \mid aAB \mid aB \\ S &\rightarrow ab \mid Ab \mid b \mid acBc \mid bbCBa \mid aACB \mid ABA \mid bc \mid aCB \mid AB \mid BA \mid bbBa \mid aAB \mid aB \\ A &\rightarrow baB \mid a \mid bac \mid CA \mid acBc \mid bbCBa \mid aACB \mid ABA \mid bc \mid aCB \mid AB \mid BA \mid bbBa \mid aAB \mid aB \\ B &\rightarrow acBc \mid bbCBa \mid aACB \mid ABA \mid bc \mid aCB \mid AB \mid BA \mid bbBa \mid aAB \mid aB \\ C &\rightarrow a \mid bac \mid CA \mid baB \mid acBc \mid bbCBa \mid aACB \mid ABA \mid bc \mid aCB \mid AB \mid BA \mid bbBa \mid aAB \mid aB \end{aligned}$$

Step 8: Create new rules that satisfies Chomsky Normal Form.

$$\begin{aligned} S_0 &\rightarrow S_a S_b \mid AS_b \mid b \mid ES_c \mid HS_a \mid JB \mid KA \mid S_b S_c \mid LB \mid AB \mid BA \mid MS_a \mid IB \mid S_a B \\ S &\rightarrow S_a S_b \mid AS_b \mid b \mid ES_c \mid HS_a \mid JB \mid KA \mid S_b S_c \mid LB \mid AB \mid BA \mid MS_a \mid IB \mid S_a B \\ A &\rightarrow NB \mid a \mid NS_c \mid CA \mid ES_c \mid HS_a \mid JB \mid KA \mid S_b S_c \mid LB \mid AB \mid BA \mid MS_a \mid IB \mid S_a B \\ B &\rightarrow ES_c \mid HS_a \mid JB \mid KA \mid S_b S_c \mid LB \mid AB \mid BA \mid MS_a \mid IB \mid S_a B \\ C &\rightarrow a \mid NS_c \mid CA \mid NB \mid ES_c \mid HS_a \mid JB \mid KA \mid S_b S_c \mid LB \mid AB \mid BA \mid MS_a \mid IB \mid S_a B \\ S_a &\rightarrow a \\ S_b &\rightarrow b \\ S_c &\rightarrow c \\ D &\rightarrow S_a S_c \\ E &\rightarrow DB \\ F &\rightarrow S_b S_b \\ G &\rightarrow FC \\ H &\rightarrow GB \\ I &\rightarrow S_a A \\ J &\rightarrow IC \\ K &\rightarrow AB \\ L &\rightarrow S_a C \\ M &\rightarrow FB \\ N &\rightarrow S_b S_a \end{aligned}$$

9. Convert the following context-free grammar into Chomsky Normal Form (CNF).

$$\begin{aligned} S &\rightarrow aSa \mid A \mid AB \\ A &\rightarrow C \mid \lambda \mid baB \mid B \\ B &\rightarrow acBc \mid bbBa \mid A \mid b \\ C &\rightarrow a \mid bac \mid CBA \end{aligned}$$

Step 1: Create new Start Variable S_0 .

$$\begin{aligned} S_0 &\rightarrow S \\ S &\rightarrow aSa \mid A \mid AB \\ A &\rightarrow C \mid \lambda \mid baB \mid B \\ B &\rightarrow acBc \mid bbBa \mid A \mid b \\ C &\rightarrow a \mid bac \mid CBA \end{aligned}$$

Step 2: Remove the λ -rules. $A \rightarrow \lambda$

$$\begin{aligned} S_0 &\rightarrow S \mid \lambda \\ S &\rightarrow aSa \mid A \mid AB \mid B \\ A &\rightarrow C \mid baB \mid B \\ B &\rightarrow acBc \mid bbBa \mid A \mid b \\ C &\rightarrow a \mid bac \mid CBA \mid CB \end{aligned}$$

Step 3: Remove Unit rules. $A \rightarrow C$ and $A \rightarrow B$

$$\begin{aligned} S_0 &\rightarrow S \mid \lambda \\ S &\rightarrow aSa \mid A \mid AB \mid B \\ A &\rightarrow baB \mid a \mid bac \mid CBA \mid CB \mid acBc \mid bbBa \mid b \\ B &\rightarrow acBc \mid bbBa \mid A \mid b \\ C &\rightarrow a \mid bac \mid CBA \mid CB \end{aligned}$$

Step 4: Remove Unit rules. $B \rightarrow A$

$$\begin{aligned} S_0 &\rightarrow S \mid \lambda \\ S &\rightarrow aSa \mid A \mid AB \mid B \\ A &\rightarrow baB \mid a \mid bac \mid CBA \mid CB \mid acBc \mid bbBa \mid b \\ B &\rightarrow acBc \mid bbBa \mid b \mid baB \mid a \mid bac \mid CBA \mid CB \\ C &\rightarrow a \mid bac \mid CBA \mid CB \end{aligned}$$

Step 5: Remove Unit rules. $S \rightarrow A$, $S \rightarrow B$, and $S_0 \rightarrow S$

$$\begin{aligned} S_0 &\rightarrow aSa \mid AB \mid baB \mid a \mid bac \mid CBA \mid CB \mid acBc \mid bbBa \mid b \mid \lambda \\ S &\rightarrow aSa \mid AB \mid baB \mid a \mid bac \mid CBA \mid CB \mid acBc \mid bbBa \mid b \\ A &\rightarrow baB \mid a \mid bac \mid CBA \mid CB \mid acBc \mid bbBa \mid b \\ B &\rightarrow acBc \mid bbBa \mid b \mid baB \mid a \mid bac \mid CBA \mid CB \\ C &\rightarrow a \mid bac \mid CBA \mid CB \end{aligned}$$

Step 6: Create new rules that satisfies Chomsky Normal Form.

$$\begin{aligned} S_0 &\rightarrow DS_a \mid AB \mid EB \mid a \mid ES_c \mid FA \mid CB \mid IS_c \mid JS_a \mid b \mid \lambda \\ S &\rightarrow DS_a \mid AB \mid EB \mid a \mid ES_c \mid FA \mid CB \mid IS_c \mid JS_a \mid b \\ A &\rightarrow EB \mid a \mid ES_c \mid FA \mid CB \mid IS_c \mid JS_a \mid b \\ B &\rightarrow IS_c \mid JS_a \mid b \mid EB \mid a \mid ES_c \mid FA \mid CB \\ C &\rightarrow a \mid ES_c \mid FA \mid CB \\ S_a &\rightarrow a \\ S_b &\rightarrow b \\ S_c &\rightarrow c \\ D &\rightarrow S_a S \\ E &\rightarrow S_b S_a \\ F &\rightarrow CB \\ G &\rightarrow S_a S_c \\ H &\rightarrow S_b S_b \\ I &\rightarrow GB \\ J &\rightarrow HB \end{aligned}$$